

Math 54 Discussion Worksheet

Name: _____

ID: _____

Problem 1

Solve the initial value problem

$$y''' - 2y'' - y' + 2y = 0$$

subject to $y(0) = 2, y'(0) = 3, y''(0) = 5$.

Problem 2

Solve the initial value problem

$$y'' - y = 0$$

first subject to $y(0) = 1, y'(0) = 0$, then subject to $y(0) = 0, y'(0) = 1$. What well known functions arise? Then, show that the two solutions are derivatives of each other.

Problem 3

Let's prove the sum of angles formula in an interesting way. Fix a point $x \in \mathbb{R}$.

- a) Define $f(t) = \sin(x+t)$. Show that $f''(t) + f(t) = 0$, $f(0) = \sin(x)$, and $f'(0) = \cos(x)$.
- b) Solve the initial value problem $y'' + y = 0$ subject to $y(0) = \sin(x)$, $y'(0) = \cos(x)$.
- c) By the uniqueness theorem, the solution in part b) is same as in part a). What do these calculations imply?